

SAMMER HUSSAIN

Full Stack AI Engineer

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SUMMARY

Full-Stack AI Engineer with hands-on experience building production-ready AI systems and SaaS products end to end. Specialized in Generative AI, LLM-powered applications, RAG pipelines, and workflow automation. Strong background in designing scalable backends, modern frontends, and intelligent automation using Python, Flask, Node.js, React, Streamlit LangChain, vector databases, and n8n. Proven ability to convert unstructured data and manual processes into reliable, scalable AI-driven solutions.

EXPERIENCE

AI Engineer Intern

CoreTech Innovations

- 08/2025 - 09/2025 Hyderabad, Sindh, Pakistan
- Developed ML models for sentiment analysis and student marks prediction using Python, Scikit-learn, and TensorFlow.
 - Built a smart service platform to streamline workflows and boost productivity.
 - Handled data preprocessing, feature engineering, model evaluation, and optimization for production-ready solutions.
 - Collaborated in Agile teams and communicated insights to drive data-informed business decisions.

EDUCATION

Bachelor of Science in Information Technology

Quaid-e-Awam University of Engineering, Sciences and Technology

12/2021 - 09/2025 Nawabshah, Pakistan

CGPA

3.58 / 4.0

SKILLS

Python	JavaScript	SQL	N8N	LangChain	RAG
Prompt Engineering	ChromaDB	Pinecone	Hugging Face		
OpenAI	Google Gemini	Transformers	Model fine-tuning		
NLTK	Scikit-learn	TensorFlow	Keras	OpenCV	
Flask	Streamlit	Node JS	Express JS	React JS	
SQLite	PostgreSQL	MongoDB	Git	GitHub	
Visual Studio Code	Anaconda	Jupyter Notebook			
Google Colab	Docker (Basic)	AWS (Basic)			

CERTIFICATION

Google AI Essentials Introduction to Generative AI
Full Stack MEAN Developer

PROJECTS

FredCoach - AI Coaching Platform

10/2025 - 11/2025

Full-Stack AI Coaching SaaS Platform

- Built a subscription-based AI coaching platform using React, Node.js, Express, and MongoDB.
- Implemented a GPT-4o chatbot with RAG, using ChromaDB for semantic document retrieval.
- Integrated Stripe subscriptions with webhook-based access control.
- Developed JWT authentication with role-based admin and user access.
- Created an admin dashboard to manage AI behavior, prompts, and branding dynamically.
- Applied backend security practices including rate limiting, CORS, and input validation.

Content Examiner - AI-Powered Plagiarism Detection System (FYP)

02/2025 - 06/2025

Final year project focused on plagiarism detection.

- Built a Flask + React web application for AI-based plagiarism detection using RoBERTa (transformers) and TF-IDF with cosine similarity.
- Implemented document and text comparison with support for PDF, DOCX, and TXT, including JWT-based authentication and secure APIs.
- Generated detailed plagiarism reports with highlighted text, source links, visual charts, and PDF export, optimized using NLTK preprocessing.

VoxMind AI - RAG-Based Voice Agent

01/2026 - 01/2026

Built an end-to-end voice & text AI agent using n8n orchestration, Google Gemini (LLM + embeddings), and Pinecone for long-term memory (RAG).

- Implemented automated document ingestion from Google Drive API, chunking, and vectorization via LangChain and Gemini embeddings.
- Enabled semantic retrieval and hallucination-resistant responses by injecting Pinecone context into Gemini prompts.
- Designed a scalable, modular webhook-based architecture using n8n for API and agent integration.

Live Object Detection with YOLOv8

08/2024 - 09/2024

Real-time full-stack computer vision application.

- Built a real-time object detection system using YOLOv8, with a Flask backend and React frontend for live camera and image uploads.
- Added distance estimation, adjustable detection settings, real-time metrics, and visual overlays for detected objects.
- Optimized the system for fast performance with real-time feedback and smooth user interaction.